

Cybersecurity Practical Report

Kali Linux Lab Manual

This document contains step-by-step solutions, commands (POC), impacts, and mitigation techniques for cybersecurity practical tasks.

1. Nmap (Network Mapper)

Aim: Perform network scanning and service detection.

Commands (POC):

```
nmap -sn 192.168.1.0/24
nmap 192.168.1.5
nmap -sV 192.168.1.5
nmap -O 192.168.1.5
nmap -A 192.168.1.5
```

Output: Open ports and OS identified.

Impact: Exposed services can be exploited.

Prevention: Use firewalls and close unused ports.

2. Wireshark

Aim: Capture and analyze network packets.

Commands (POC):

```
icmp
tcp.port == 443
dns
```

Output: Packets captured and analyzed.

Impact: Sensitive data interception possible.

Prevention: Use encryption (HTTPS).

3. John the Ripper

Aim: Crack password hashes.

Commands (POC):

```
john --wordlist=/usr/share/wordlists/rockyou.txt hashes.txt
john --show hashes.txt
```

Output: Passwords cracked.

Impact: Weak passwords compromised.

Prevention: Use strong passwords and hashing.

4. WHOIS

Aim: Retrieve domain information.

Commands (POC):

```
whois google.com
```

Output: Registrar and DNS info found.

Impact: Information disclosure risk.

Prevention: Use WHOIS privacy.

5. DIG

Aim: Retrieve DNS records.

Commands (POC):

```
dig google.com
dig google.com MX
dig google.com NS
```

Output: DNS records listed.

Impact: Infrastructure mapping.

Prevention: Restrict DNS exposure.

6. TheHarvester

Aim: Collect emails and subdomains.

Commands (POC):

```
theHarvester -d example.com -b all
```

Output: Emails and subdomains found.

Impact: Phishing risk.

Prevention: Limit public exposure.

7. Zphisher

Aim: Perform phishing attack.

Commands (POC):

```
git clone https://github.com/htr-tech/zphisher
cd zphisher
bash zphisher.sh
```

Output: Fake login page created.

Impact: Credential theft.

Prevention: User awareness, 2FA.

8. Burp Suite Intruder

Aim: Brute-force login credentials.

Steps:

- Intercept request
- Send to Intruder
- Add payload
- Start attack

Impact: Account compromise.

Prevention: CAPTCHA, rate limiting.

9. Sublist3r

Aim: Find subdomains.

Commands (POC):

```
sublist3r -d vulnweb.com -o output.txt
```

Output: Subdomains listed.

Impact: Expanded attack surface.

Prevention: Secure subdomains.

10. Dirbuster

Aim: Discover hidden directories.

Commands (POC):

```
dirbuster
```

Output: Hidden directories found.

Impact: Sensitive data exposure.

Prevention: Proper access control.

11. Shodan

Aim: Search exposed devices.

Commands (POC):

```
shodan search apache
```

Output: Devices listed.

Impact: Exposed services.

Prevention: Secure configurations.

12. Hydra

Aim: Perform brute-force attack.

Commands (POC):

```
hydra -l admin -P rockyou.txt ssh://192.168.1.5
```

Output: Credentials found.

Impact: Unauthorized access.

Prevention: Strong passwords.

13. DNSenum

Aim: Gather DNS information.

Commands (POC):

```
dnsenum google.com
```

Output: DNS records.

Impact: Network mapping.

Prevention: Limit DNS info.

14. Gobuster

Aim: Brute-force directories.

Commands (POC):

```
gobuster dir -u http://testphp.vulnweb.com -w common.txt
```

Output: Hidden files found.

Impact: Exposed endpoints.

Prevention: Disable directory listing.

15–24. DVWA & Metasploitable Attacks

Aim: Perform various attacks (XSS, SQLi, File Inclusion, etc.).

Commands (POC):

```
<script>alert("XSS")</script>
' OR 1=1 --
; ls
../../../../etc/passwd
```

Output: Vulnerabilities exploited.

Impact: Full system compromise possible.

Prevention: Input validation, patching, security controls.